

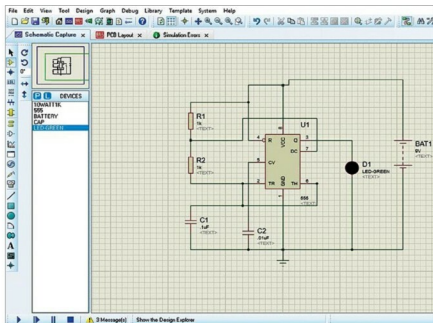


Fig. 21: Schematic window

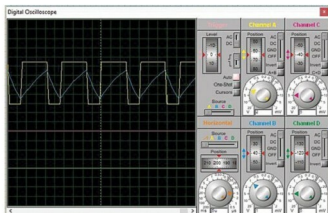


Fig. 22: Oscilloscope output

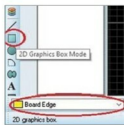


Fig. 23: Selecting Board Edge option

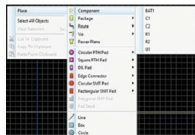


Fig. 24: Placing the components

release selected device.

5. After placing components over the schematic, connect their terminals by clicking-and-dragging technique. Moving the mouse cursor to the terminals will automatically show a pen symbol for drawing

(Fig. 21).

6. Ensure all connections are made and save the project before simulation. To run the circuit, go to Debug→Run Simulation or simply press F12. If you connect an oscilloscope to the circuit while running

the simulation, a new digital oscilloscope window will pop up and show the waveforms (Fig. 22).

7. After successful simulation of the circuit, move to the PCB layout. Before going to the PCB design, ensure that all components have their own footprint. Like all other EDA tools, Proteus also has a different tool for PCB design, which is an integrated ARES PCB designing suite. Select Netlist option to ARES from Tools menu; it will open a new PCB layout design window.

8. Create a board edge. Select 2D Graphics Box Mode in the left corner and select Board Edge from the dropdown list in the left-most corner (Fig. 23).

9. On the workspace, draw a box of sufficient size in which you can place all the components. The circuit should be inside the yellow box; if circuit is complex, you can change the size of the box later.

10. Place each component outside the yellow box by right-clicking option Place→Component (Fig. 24).

11. After adding all components to the workspace, arrange the positions of the components properly in the workspace.

12. Next step is tracking, which establishes the connection between components. Connection representation is in green line; yellow line shows direction. After setting track width, click on one end of the component with pen and follow the green line. When two components are successfully connected, the green line will automatically disappear. Multiple PCB layers are also possible in Proteus.

13. In Tool menu, options like auto router and auto placer can also be used by beginners, but manual routing and placing is recommended for better results.

14. To see a 3D image of the final circuit, click on Option 3D Visualization. The PCB layout can be printed from Print Layout option in Output menu. **EFY**